

Esteban Sitek

Sr. Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **12+** years across fintech, healthcare, and e-commerce, building scalable data pipelines and warehousing solutions using Python, Spark, and cloud SQL platforms. Expert in ETL orchestration, data quality frameworks, and analytics infrastructure using Apache Spark, BigQuery, and Airflow. Committed to delivering production-grade data systems that enable business intelligence and predictive analytics.

SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- Data & ML:** Apache Spark, Airflow, Pandas, NumPy, scikit-learn, dbt, Kafka
- Backend:** Node.js, FastAPI, Django, Spring Boot, RabbitMQ, gRPC
- Database:** Google BigQuery, PostgreSQL, MySQL, AWS Redshift, Redis, Elasticsearch
- Testing:** Pytest, Jest, Selenium, JUnit
- Cloud & DevOps:** Google Cloud Platform, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- Others:** RESTful API, GraphQL, Grafana, Prometheus, Datadog, Git, Jira, Agile

EXPERIENCE

Publicis Sapient Jun 2024–Present
Senior Software Engineer

- Led the architecture and implementation of real-time data ingestion pipelines using **Python** and **Apache Spark**, processing multi-terabyte datasets from heterogeneous enterprise systems across financial services and healthcare verticals.
- Designed and deployed **BigQuery**-backed analytics infrastructure with automated data quality checks using **Airflow**, reducing data validation errors by **42%** and accelerating report delivery from days to hours.
- Collaborated with analytics and business intelligence teams to define KPI metrics and construct visualization dashboards in **Grafana**, enabling real-time operational insights for stakeholders across five business domains.
- Architected a scalable Data Warehouse schema using dimensional modeling and **dbt** transformations, supporting 150+ downstream analytics queries and machine learning feature engineering pipelines.
- Mentored three junior data engineers on distributed computing concepts, **Apache Spark** optimization, and production data pipeline best practices, raising team capability and code review quality across the platform.

- Participated in incident resolution when a data pipeline stalled due to malformed API responses — added defensive parsing logic and schema validation in **Python**, preventing data loss and ensuring 99.8% pipeline uptime.
- Implemented incremental ETL workflows using **Kafka** event streams, cutting batch processing latency from 4 hours to 22 minutes and enabling near-real-time analytics for fraud detection features.
- Extended the data infrastructure to support multi-country operations, building region-specific **PostgreSQL** replicas and **Terraform** automation to provision isolated environments for data residency compliance.
- Refined deployment processes by containerizing data jobs with **Docker** and orchestrating them on **Kubernetes**, reducing environment inconsistencies and cutting job failure rates by **35%**.
- Collaborated with the platform engineering team to integrate **Prometheus** metrics and alerting into all critical data workflows, improving observability and reducing incident detection time by **55%**.
- Drove adoption of **Git** workflows and peer review standards across the data team, reducing regressions in production pipelines by establishing automated **Pytest** coverage thresholds.
- Evaluated and piloted **AWS Redshift** as an alternative to BigQuery for specific workloads, providing cost-benefit analysis that improved query performance by **28%** for large-scale historical analytics.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Spearheaded the migration of patient monitoring data pipelines from legacy batch jobs to a **Kafka**-driven architecture using **Python** and **Apache Spark**, handling 8 million daily events with sub-second latency.
- Owned the design and deployment of a **BigQuery** data lake for healthcare analytics, consolidating patient records, clinical outcomes, and remote monitoring telemetry from 200+ hospital systems.
- Built **dbt** transformation models and SQL views to prepare denormalized datasets for hospital reporting dashboards, reducing report generation time from 3 hours to 8 minutes and enabling real-time clinical decision support.
- Implemented a data quality framework in **Python** with automated anomaly detection, flagging missing vital signs, erroneous timestamps, and duplicate records before they reached downstream systems.
- Partnered with clinical operations to define population health metrics and KPIs, building interactive **Grafana** dashboards that tracked readmission rates, intervention success, and resource utilization across care programs.
- Refactored legacy **MySQL** queries that were timing out on large patient cohorts, optimizing indexes and rewriting inefficient joins, cutting query time from 45 seconds to under 2 seconds.
- Led cross-functional work to enrich healthcare data with external sources including lab results and pharmacy records, using **Node.js** API adapters and **FastAPI** microservices to normalize and validate third-party data feeds.
- Reduced on-call incidents related to data pipeline failures by implementing **Datadog** monitoring and **Airflow** auto-retry logic with exponential backoff for transient failures.
- Collaborated with the machine learning team to prepare training datasets for predictive readmission models, creating reproducible feature engineering pipelines that **scikit-learn** models consumed directly.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Drove the development of transactional data pipelines handling 50 million daily e-commerce events using **Apache Spark** and **Python**, ingesting order, payment, and logistics data from 6 Southeast Asian markets.
- Built a distributed **PostgreSQL** sharding strategy to partition customer order history by geography, improving query response times by **31%** and enabling localized analytics for regional marketing teams.
- Owned the ETL workflows that transformed raw clickstream and inventory data into **BigQuery** datasets, powering recommendation engines and merchant analytics that increased seller engagement by 18% quarter-over-quarter.
- Implemented a schema versioning system using **dbt** to manage evolving data models across 40+ data sources, preventing breaking changes and reducing data contract disputes between teams.
- Designed **Grafana** dashboards tracking fulfillment metrics, payment success rates, and fraud signals, providing product and operations teams with real-time visibility into platform health across 15+ business KPIs.
- Optimized slow-running **SQL** queries on order history tables by analyzing execution plans and adding strategic **Redis** caching layers, cutting report generation latency by **52%** during peak traffic.
- Collaborated with backend engineers to standardize **RESTful API** contracts and payload schemas for data feeds, ensuring consistency and reducing downstream transformation complexity across the data warehouse.
- Participated in a data recovery incident after a Spark job corrupted a critical analytics table — implemented defensive checksums and rollback procedures in **Python**, establishing a recovery SLA of under 30 minutes.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built server-side analytics pipelines for free-to-play games, collecting and processing player event logs from 5 million monthly active users using **Python** and **Node.js** backend services.
- Designed a **MySQL** database schema for game telemetry covering player progression, in-game economy, and engagement metrics, optimizing queries to support real-time dashboards viewed by the product team.
- Implemented **Elasticsearch**-backed search and aggregation for player analytics, enabling the data team to quickly analyze cohort behaviors and identify engagement trends across game titles.
- Created automated reporting pipelines that extracted game metrics into **SQL** views and visualization tools, providing stakeholders with weekly performance summaries and debugging historical data issues.
- Contributed to early-stage work on data infrastructure by building **JavaScript** tools for data validation and quality checks, catching malformed event payloads before they entered the analytics pipeline.
- Helped establish logging standards and centralized log collection, laying groundwork for future monitoring and observability practices as the studio scaled game deployments.

EDUCATION

Ateneo de Manila University
Bachelor of Science in Computer Science

Aug 2010–Jul 2014